EXTENDED ABSTRACT

Undergraduate Thesis/Master Thesis/Dissertation **(delete the irrelevant option)**

Title of your abstract, capitalize first letter and proper nouns - **(Font: Times New Roman, 14, Bold, cantered, Line Space 1.5)**

Student Name: (Font: Times New Roman, 12, centered, Bold, Line Space 1)
Student ID Number (NPM): (Font: Times New Roman, 12, centered, Line Space 1)
Study Program: (Font: Times New Roman, 12, centered, Line Space 1)

Body/Text of the abstract: 1750 words and maximum 4 pages, with a few subheadings (Font: Times New Roman, 12, justified, Line Space 1.15)

(Paragraph 1 Research Background, Objectives, and Methodology)

Background and Novelty: Briefly state the research gap and the urgency of the study. Explicitly mention the **Novelty** here (e.g., "This study provides a new approach to..." or "Unlike previous models, this research introduces...").

Objectives: State the primary goal of the dissertation clearly (e.g., "The objective is to evaluate the impact of...").

Methodology & Schematic Diagram: Describe the research design and data analysis briefly. To ensure the logic is understood by English speakers, the following schematic diagram illustrates the complete workflow:

Figure 1. Schematic Diagram of Research Framework

[Insert Diagram Here]

(Paragraph 2-4 Results and Discussion), Results and Discussion (70% of the Extended Abstract)

(Provide a detailed, multi-paragraph technical explanation. This is the primary section for readers to understand the substance of your Indonesian-language dissertation.)

Paragraph 2: [Specific Finding Name] Detail the primary data obtained. Do not just state the result; explain the technical significance.

Paragraph 3: [Analysis and Interpretation] Explain the "why" behind your results. Compare your findings extensively with international literature. Discuss how your data confirms or challenges existing global theories.

Paragraph 4: [Scientific Implications] Describe the practical and theoretical impact of your findings. Use this space to demonstrate the depth of your dissertation's contribution to the field.

A maximum of 2 figures and/or tables may be included in this section.

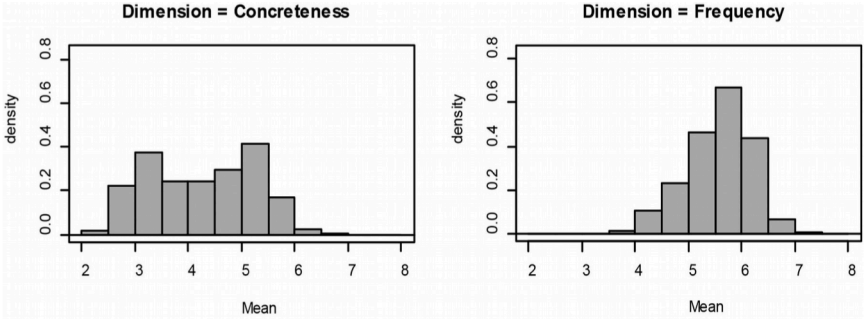


Figure 2. Figure Header
Reference: Author surname and publication date (e.g., Widakdo, 2022)

Table 1. Summary of rating dimensions with means (Mean), standard deviation.
Reference: Author surname and publication date (e.g., Widakdo, 2022)

Dimension	Mean	SD	Range	Median
Valence	4.97	1.79	2.08-7.92	5.20
Arousal	4.86	2.04	3.04-7.28	4.79
Dominance	5.25	2.09	3.74-6.80	5.23
Unpredictability	5.24	2.27	2.83-7.16	5.24
Concreteness	4.27	2.21	2.31-6.84	4.27
Frequency	5.57	2.21	3.21-7.39	5.64

(Paragraph 5 Conclusion)
Summarize the main conclusions of the dissertation and highlight the broader impact or future potential of the research.

(Paragraph 6 Keywords)
Keywords: (3-5 keywords, arranged alphabetically, separated by commas, Font: Times New Roman, 12, Left Aligned)

Example of extended abstract (dummy)

EXTENDED ABSTRACT
Undergraduate Thesis

Unlocking Piezoelectric Potential of PVDF/Graphene Coatings on Melamine Sponges for Sensor Technologies

Student Name: Januar Widakdo
Student ID Number (NPM): 1234567890
Study Program: Undergraduate Physics

Sensors play a vital role in advancing technology, particularly in healthcare and electronic devices. However, conventional piezoelectric materials like lead-based ceramics (PZT) are rigid, brittle, and toxic, which limits their application in wearable technologies . This research introduces a novel approach by utilizing melamine sponges, which possess inherent piezoelectric properties, as the core material (see **Figure 1**). The novelty lies in integrating these sponges with polyvinylidene fluoride (PVDF) and graphene to enhance their functionality for next-generation flexible sensors. The objective is to evaluate the impact of graphene concentration on the piezoelectric effect and sensor performance. The methodology involves fabricating PVDF and PVDF-graphene-coated melamine sponges using a dip-coating method with concentrations ranging from 6 wt% to 8 wt%. The samples were dried at 100°C for 12 hours. Characterization was performed using Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), and Attenuated Total Reflectance-Fourier Transform Infrared Spectroscopy (ATR-FTIR), followed by piezoelectric testing under mechanical loads of 1000–3000 g

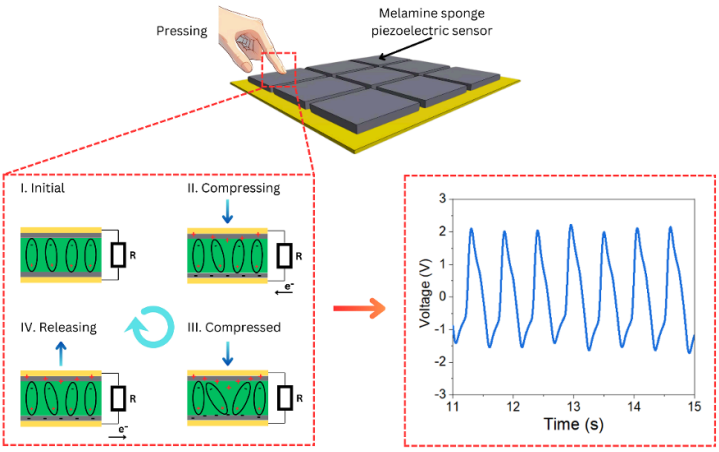


Figure 1. Schematic Diagram of Research Framework

The experimental results demonstrate that the 8% PVDF-Gr sample achieved the most optimal performance. ATR-FTIR analysis indicated that the β-phase ratio increased significantly from 19% in pure PVDF to 64% after graphene addition . This was further confirmed by XRD data, which showed that the 8% PVDF-Gr sample contained 79.34% of the electroactive β-phase. The electrical performance of these samples is illustrated in **Figure 2**, which displays the Vpp output under various mechanical loads. While the pure 8% PVDF sample generated only 0.29 V under a 3000 g load, the 8% PVDF-Gr sample produced a significant voltage of 3.55 V. This drastic increase is attributed to the conductive network formed by graphene's delocalized π-electrons, which facilitates efficient charge transfer .

The addition of graphene triggers a molecular transformation in PVDF from the non-electroactive α-phase to the electroactive β-phase. This occurs because the net surface charges of graphene interact with the dipoles of PVDF monomers (CH_2 and CF_2), inducing an all-trans molecular conformation. The porous and highly compressible nature of the melamine sponge substrate further enhances this effect by increasing the amount of strain generated under pressure, leading to a more pronounced electrical response. When compared to international literature, our 8% PVDF-Gr system (V_{pp} = 3.55 V) outperforms other configurations such as mica-based PZT sensors (1.8 V) or PVDF-TrFE footstep sensors (1 V), demonstrating its superior energy-harvesting and sensing capabilities.

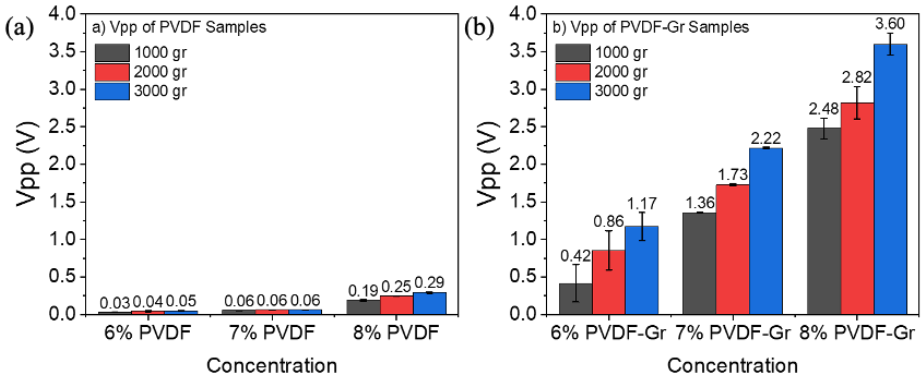


Figure 2. Vpp values of (a) PVDF samples and (b) PVDF-Gr samples under varying mechanical loads

The interaction between the net surface charges of graphene and the dipoles of PVDF monomers (CH_2 and CF_2) induces an all-trans molecular conformation, effectively transforming the non-electroactive α-phase into the β-phase . The porous structure of the melamine sponge enhances this effect by increasing the strain generated under pressure. As summarized in **Table 1**, the 8% PVDF-Gr system outperforms several configurations found in international literature, such as mica-based PZT sensors and PVDF-TrFE footstep sensors. These findings suggest that the material can be effectively used to detect

human motoric movements, such as walking, by monitoring voltage peaks corresponding to each step.

Table 1. Comparison of Vpp between this study and prior piezoelectric sensor research.

Sample	Application	Vpp (V)
Aligned electrospun PVDF/CNT	Wearable sensor	0.8
Neat PVDF	Pressure sensor	1.2
PVDF-TrFE	Footstep sensor	1
8% PVDF-Gr	**Motoric movement**	**3.55**

In conclusion, the addition of graphene significantly enhances the piezoelectric properties of PVDF by promoting the formation of the β-phase up to 79.34%. The 8% PVDF-Gr sample emerged as the most optimal condition, producing a voltage output more than seven times higher than pristine PVDF. These findings highlight the critical role of graphene in improving the performance of flexible composite materials for future sensor technologies

Keywords: Graphene, Melamine sponge, Piezoelectric, PVDF